



CHANDU BAYA REDDY

M.Sc. Information & Communication Technology

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Portfolio Website

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About

M.Sc. ICT student at FAU Erlangen specializing in Networks & Digital Communication. Brings hands-on industrial experience from **Robert Bosch** in automating automotive sensor validation and conducting root-cause failure analysis. Currently starting a **Master's Thesis** on Differential TDoA-based BLE localization at **Fraunhofer IIS**, in collaboration with the Chair of Information Technology and Communication Electronics (LIKE), FAU Erlangen-Nürnberg. Separately, actively seeking a Working Student role in IoT, Embedded Systems, or Wireless Communication (available 20h/week during the semester).

Education

FAU Erlangen-Nuremberg

Master of Science in Information and Communication Technology

Major: Networks & Digital Communication

April 2024 – current

Erlangen, Bayern, Germany

Reva University, Bengaluru

Bachelor of Technology in Electronics and Communication

Focus: Digital Signal Processing, Wireless Communications

Grade: First Class with Distinction (GPA 1.4 Equivalent)

August 2018 – June 2022

Bengaluru, Karnataka, India

Current Projects

mioty® & BLE Hybrid Localization Tag (Ultra-Low Power Design)

Aug 2025 – Aug 2026

Tech Stack: mioty (TSMA), BLE, Google FMDN, STM32F103, SX1280, KiCad, Leaflet.js, Power Budgeting

- **Architected a dual-protocol tracking system**, integrating mioty for long-range telemetry and BLE for Google's Find My Device Network (FMDN) ecosystem.
- **Optimized system power consumption** for standard CR2032 coin cells by implementing aggressive firmware sleep states and hardware reservoir capacitors to handle peak transmission bursts.
- **Validated end-to-end in the field:** a real Android device resolved the tag's position via Google's FMDN network, which typically resolves to within ~10m as more devices observe the tag, with dual-protocol operation projecting ~7.5 months of battery life (~41 μ A avg. current) on a single CR2032.

Audio Signal Processing & Binaural Synthesis

April 2025 – July 2025

Tech Stack: Python, Jupyter, Librosa, SciPy/NumPy, Matplotlib, LaTeX

- Implemented foundational DSP algorithms from scratch, including **STFT, convolution operations, and Pitch Class Profiles** for speech and spectral analysis.
- Researched and presented state-of-the-art neural audio codecs, delivering a comprehensive seminar on End-to-End Binaural Speech Synthesis and adversarial training architectures.

Professional Experience

Fraunhofer IIS

Master's Thesis Student

Oct 2026 – Ongoing

Erlangen, Germany

Differential TDoA Localization Using Bluetooth Signals

Tech Stack: BLE, TDoA, SDR (USRP), Differential Correlation, Synchronization Analysis, GNU Radio

- Investigating the adaptation of an existing differential TDoA localization algorithm, originally developed for the mioty LPWAN standard, to Bluetooth Low Energy signals, in collaboration with the Chair of Information Technology and Communication Electronics (LIKE), FAU Erlangen-Nürnberg.
- Analyzing the influence of anchor-node synchronization errors on achievable positioning accuracy, and implementing/evaluating the system using a multi-anchor Software-Defined Radio (SDR) setup.

Product Development | Sensors | Power-Train Solutions

Tech Stack: Raspberry Pi, Arduino, Python (NumPy, pandas, PyQt5), MATLAB/Simulink, CAN/SENT, Linux, EasyEDA, JIRA.

- Designed and validated **sensor test control units**; built automated measurement routines in Python and MATLAB to acquire and analyze signals, reducing test time by 40% and operational costs by 50%.
- Performed **electrical measurements** on automotive powertrain sensors (O2, NOx, pressure, and throttle position) via CAN/SENT interfaces, executed calibration of test benches, and documented **traceable testing procedures**.
- Led failure analysis and **root-cause investigations** for noisy/unstable signals (clipping, drift, wiring faults) using spectral analysis, comprehensive logging, and reproducible test scripts.
- Produced comprehensive release-readiness reports containing data plots, strict acceptance criteria, and equipment calibration records.

Technical Skills

Wireless & IoT: mioty (TSMA/ETSI), BLE, 5G/6G Topologies, LPWAN, CAN/SENT, MQTT, I2C, SPI, UART.

Embedded Systems: STM32, Arduino, Raspberry Pi, Embedded C, Bare-metal Programming, FreeRTOS, Python, FPGA Architectures, Power Budgeting.

Hardware Design: KiCad, Altium, Multilayer PCB Design, Signal Integrity, LTspice.

Hardware Debugging: Oscilloscopes, Logic Analyzers, Power Analysers, Spectrum Analysers, JTAG/SWD.

Analysis & Tools: STM32CubeIDE, MATLAB, Simulink, PyTorch, Pytest, Git, Linux (Ubuntu), JIRA, PowerBI, LaTeX.

Methodologies & Testing: Hardware-in-the-Loop (HIL), Agile/Scrum methodologies.

Projects (Bachelors)

Solar Powered Wheelchair with Line Follower system with RF-ID Authentication

Nov 2021 – April 2022

Tech Stack: Arduino-Uno, RC-522, IR sensor array, Motor-driver L298N, Relay 10A/5V, ESP32-cam, Solar panels.

- Developed autonomous navigation wheelchair powered by solar energy with RFID-based route authentication.
- Integrated line-following and obstacle avoidance system with live video streaming.

Hand Gesture controlled robot car with ESP32Cam module

April 2020 – August 2020

Tech Stack: Arduino-nano, Lilypad arduino, MPU6050 (Gyroscope& Accelerometer), Motor-drivers, Wi-Fi

- Implemented motion-based control via MPU6050 sensors, enabling intuitive navigation.
- Integrated live-streaming camera for remote control and monitoring.

Participations & Publications

- Competed in the E-yantra Robotics Competition organized by **IIT Bombay**.
- Developed a Smart-Irrigation mini-project focused on real-time analysis of IoT sensors, hosted by the **Smart India Hackathon**.
- Published a scientific paper on my Bachelor's major project, "Solar Powered Wheelchair with Line Follower system with RFID Authentication" (Article ID: IJERTCONV10IS11104). **IJERT**.

Language Skills

Languages: English (C1 - Advanced), German (A2 - Conversational), Kannada (Native), Telugu (Native), Hindi (Fluent)

Accomplishments

Awards: Employee of the Month (Robert Bosch) – for contributions to Embedded Systems & Sensor Testing in Powertrain Solutions.

Hobbies

Badminton
Cooking
Hiking